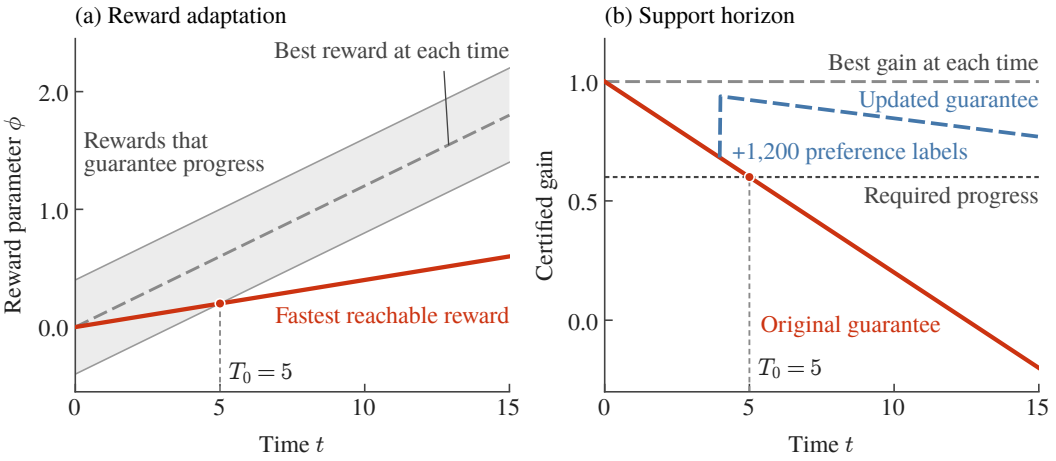
(a) Reward adaptation
Best reward at each time
Rewards that
guarantee progress
Fastest reachable reward
$T_0 = 5$
Reward parameter ϕ
2.0
1.0
0.0
0
5
10
15
Time t
(b) Support horizon
Best gain at each time
Updated guarantee
+1,200 preference labels
Required progress
Original guarantee
$T_0 = 5$
Certified gain
1.0
0.5
0.0
0
5
10
15
Time t